End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.Tech (AIML)>, Semester:<II>
(May, 2024)

Subject Code: BAM-102		Subject: Fundamentals of Data Structures	
Time :3 Hours			Maximum Marks :60
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.			
Q1		(2.5*8=20)	
	(a)	What is Data Structures ? Why do we need Data Structures ?	
	(b)	Differentiate between linear and non-linear data structure.	
	(c)	What are the advantages of Linked List, also list down its applications.	
	(d)	Define Stack ? Write down its application and operations performed on stack.	
	(e)	What is threaded binary tree ? What are the advantages of threaded binary tree over binary tree.	
	(f)	Define Hashing ? what is the need of Hashing ?	
	(g)	Define Graph data structure ? Describe different types of graphs and their applications.	
	(h)	Define different operations of graphs and their examples.	
		UNIT-I	
Q2	a)	Define an Algorithm ? List out and discuss the sequence of steps needed to design and analyze an algorithm.	(5,5)
	b)	What are the merits and demerits of array implementation of lists?	
Q3	a)	Write a program for array implementation of lists.	(5,5)
	b)	Write a program for deletion of element of array list	
		UNIT-II	
Q4	a)	a) What is priority queue ? Implement using linked lists.	(5,5)
	b)	Implement stack using linked lists.	
Q5	a)	Explain selection sort with example, also give its complexity ?	(5,5)
	b)	Explain binary search with example, also give its complexity ?	
		UNIT-III	
Q6	a)	Explain Binary search tree with example.	(5,5)
	b)	Convert the infix (a+b)*(c+d)/f into postfix & prefix expression	
Q7	a)	Construct an expression tree for the expression (a + b * c) + ((d * e + 1) * g). Give the outputs when you apply preorder, inorder and postorder traversals.	(5,5)
	b)	Explain AVL tree ADT in detail.	
		UNIT-IV	
Q8	a)	What is spanning tree? Explain how minimal spanning trees are constructed ?	(5,5)
	b)	Explain Prim's algorithm with an example?	
Q9	a)	Explain Krushal's algorithm with an example?	(5,5)
	b)	Explain different types of files organization with example.	